DIVISIBILITY AND COPRIMALITY IN LARGE SUBSETS OF $\{1, \dots, 2n\}$

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Abstract. Two classical pigeonhole facts state that any $n+1$ numbers chosen from $\{1, \dots, 2n\}$ contain two coprime numbers, and contain two numbers one of which divides the other. We give short proofs of both, show that they are sharp, and compare the second with the Erdős–Szekeres theorem on monotone subsequences.

1. Introduction

The divisibility statement (Theorem 2.5) is a well-known problem of Erdős; it is discussed together with several other pigeonhole arguments in [1]. We give a self-contained treatment in which the role of the odd part of an integer is made explicit, and we record the matching extremal examples (Propositions 3.1 and 3.2).

2. Definitions and the two pigeonhole theorems

Definition 2.1. For an integer $m \geq 0$ we write $[m] = \{1, 2, \dots, m\}$. A set A of positive integers is a *divisibility antichain* if there are no $a, b \in A$ with $a \mid b$, and a *divisibility chain* if for all $a, b \in A$ we have $a \mid b$ or $b \mid a$.

Lemma 2.2. *For every integer $k \geq 1$ we have* $\gcd(k, k+1) = 1$.

Proof. Any common divisor of k and $k+1$ divides $(k+1) - k = 1$. □

Proposition 2.3. *Let $n \geq 1$ and let $A \subseteq [2n]$ with $|A| \geq n$. Then A contains two distinct elements that are coprime.*

Proof. Partition $[2n]$ into the n pairs $\{1, 2\}, \{3, 4\}, \dots, \{2n-1, 2n\}$. Since the $n+1$ (or more) elements of A are distributed among n pairs, two of them lie in the same pair, i.e. they are consecutive integers $k, k+1$, and these are coprime by Lemma 2.2. □

Lemma 2.4. *Every integer $m \geq 1$ can be written uniquely as $m = 2^k q$ with $k \geq 0$ and q odd. We call q the* odd part *of m.*

Proof. Take k to be the exponent of 2 in the prime factorisation of m, and $q = m/2^k$. Uniqueness follows from uniqueness of prime factorisation. □

Theorem 2.5 (Main theorem)**.** *Let $n \geq 1$ and let $A \subseteq [2n]$ with $|A| \geq n+1$. Then there are distinct $a, b \in A$ with $a \mid b$.*

Proof. The odd part of an element of $[2n]$ is an odd number in $[2n]$, so it lies in $\{1, 3, \dots, 2n-1\}$, a set of n elements. By the pigeonhole principle two distinct elements of A have the same odd part q, say $a = 2^i q$ and $b = 2^j q$ with $i \neq j$; we may assume $i < j$. Then $b \mid a$, since $a = 2^{i-j} b$. □

3. Sharpness

Proposition 3.1. *For every $n \geq 1$ the set $\{n+1, n+2, \dots, 2n\} \subseteq [2n]$ is a divisibility antichain with n elements. Hence the bound $n+1$ in Theorem 2.5 cannot be lowered.*

Proof. If $a \neq b$ are in the set and $a \mid b$, then $b \geq 2a \geq 2(n+1) > 2n$, which is impossible. □

Proposition 3.2. *For every $n \geq 1$ the set $\{2, 4, \dots, 2n\} \subseteq [2n]$ has n elements, no two of which are coprime.*

Proof. Any two of its elements have the common divisor 2. □

Lemma 3.3. *Let $n \geq 1$ and let $A \subseteq [2n]$ contain no two distinct coprime elements. Then $|A| \leq n$, and every element of A is even.*

Proof. The bound $|A| \leq n$ is the contrapositive of Proposition 2.3. If $a \in A$ were odd, then a would be coprime to both $a-1$ and $a+1$ by Lemma 2.2, and by maximality one of these lies in A, a contradiction. □

Remark. Theorem 2.5 says that $[2n]$ contains no divisibility antichain of size $n+1$. It is instructive to compare this with the theorem of Erdős and Szekeres [2]: *every sequence of n^2 distinct real numbers contains a monotone subsequence of length $n+1$.* Both are consequences of the pigeonhole principle applied to a suitable labelling.

References

[1] M. Aigner and G. M. Ziegler, *Proofs from THE BOOK*, 6th ed., Springer, 2018.

[2] P. Erdős and G. Szekeres, *A combinatorial problem in geometry*, Compositio Math. **2** (1935), 463–470.

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